

INDIAN INSTITUTE OF TECHNOLOGY GUWAHATI

ESTATES AND WORKS SECTION • ROADS & CIVIL INFRASTRUCTURE DIVISION
Predictive Pavement Management System — Official Work Order

Work Order ID: WO-IITG-CL011-20260926	Date Issued: 26 September 2026, 11:16 IST
Target Route: Faculty Gate Main Road	Action Type: Full Segment Resurfacing
Center GPS: 26.19376, 91.70221	Defect Cluster Count: 1 detected defects

1. PREDICTIVE ENGINEERING & RISK ASSESSMENT

This 100-meter corridor was clustered via Spatial DBSCAN combining multi-modal inputs (proactive CCTV traffic telemetry, 100 Hz bicycle accelerometer vertical shocks & crowdsourced reports). The aggregate **Road Damage Index (RDI) is 0.514** with a cumulative surface defect area of **2.31 m²**. Compounded by pavement aging, the **Predictive Degradation Index (PDI) has reached 1.660**, crossing the critical threshold for sub-base deterioration. Reactive point patching is hereby discontinued for this segment in favor of predictive 100m cold-milling and asphaltic concrete resurfacing.

2. LIFECYCLE COST SAVINGS ANALYSIS (GRANICA COMPLIANT)

Metric / Parameter	Reactive Spot Patching	Predictive Full Resurfacing
Immediate Repair Cost	Rs. 7,500.00	Rs. 34,000.00
Expected Longevity	6 - 9 Months (High monsoon failure)	4 - 5 Years (Durable wear course)
Projected 3-Year Maintenance	Rs. 18,750.00	Rs. 34,000.00
Net Campus Savings	Baseline (0% Savings)	38.0% Cost Reduction

3. SEGMENT DEFECT MANIFEST

Observation ID	Source	Area (m²)	Peak Shock	PDI Score
OBS_IITG_0112	CCTV	2.31	2.11g	1.660

4. AUTHORIZATION & SIGN-OFF

Recommended By:
Lead Data Scientist / AI Vision Engine
IIT Guwahati Estates Automation

Approved By:
Executive Engineer (Civil)
Estates & Works Section, IIT Guwahati